

Feedback System Performance-II

These slides are based on materials from the following books:

- J. Van de Vegte. *Feedback Control Systems*. Prentice-Hall, 1986.
- W. Bolton. *Mechatronics: A Multidisciplinary Approach*. 4th Edition, Pearson, 2008.

Objectives

When you have finished this lecture you should be able to:

- Understand the motivation of the use of feedback in mechatronic systems and its effect on performance.

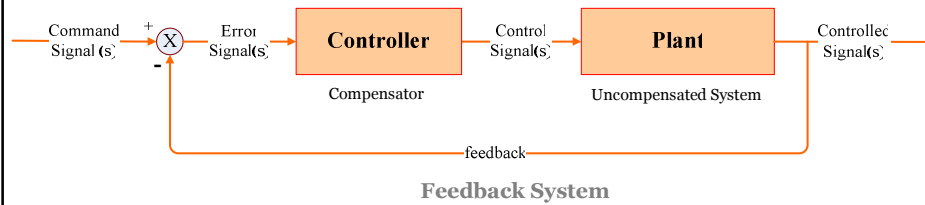
Outline

- Introduction
- Transient Response and Stability
- Transient Response vs Steady-State Errors
- Summary

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- **Introduction**
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Introduction



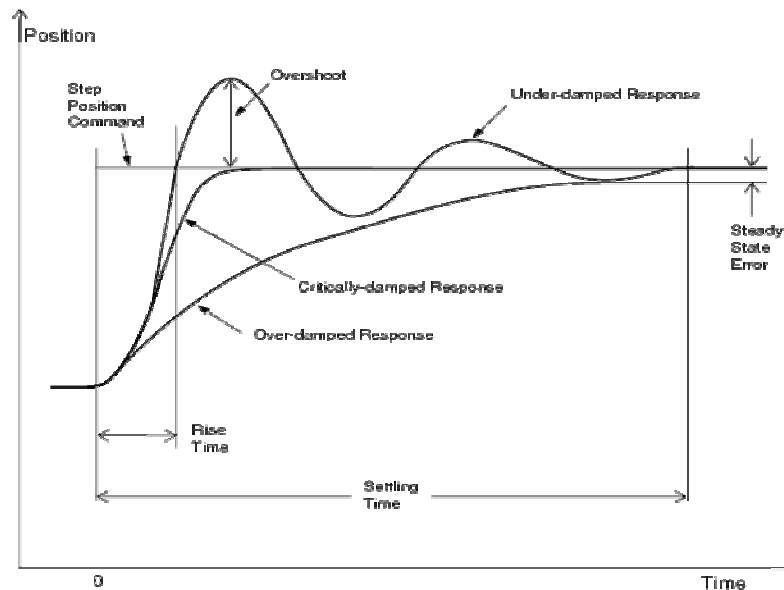
Motivations for feedback:

- Reducing the effects of parameter variations;
 - Reducing the effects of disturbance inputs;
 - Improving transient response characteristics;
 - Reducing steady-state errors.
- } Prime reasons why feedback is needed

Outline

- Introduction
- **Transient Response and Stability**
- Transient Response vs Steady-State Errors
- Summary

Transient Response and Stability



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Transient Response and Stability

System Stability is the most important characteristic of the transient response.

A system is **stable** if the transient solution **decays** to zero and is **unstable** if this solution **grows**.

A stable system is one that produces a **bounded, or finite, response** when subjected to a **bounded input**.

Stability Theorem: A system is stable if and only if all system poles lie in the left half of the s-plane.

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Transient Response and Stability

Stability Theorem: A system is stable if and only if all system poles lie in the left half of the s-plane.

The transfer function provides a basis for determining important system response characteristics without solving the complete differential equation. The transfer function is a rational function in the complex variable $s = \sigma + j\omega$, that is:

$$H(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}$$

Polynomial form

$$H(s) = \frac{\text{numerator}(s)}{\text{denominator}(s)} = \frac{N(s)}{D(s)} = K \frac{(s - z_1)(s - z_2) \dots (s - z_{m-1})(s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_{n-1})(s - p_n)}$$

$$K = b_m / a_n.$$

Expanded form of the Polynomials

Transient Response and Stability

• Poles and Zeros

$$H(s) = \frac{N(s)}{D(s)} = K \frac{(s - z_1)(s - z_2) \dots (s - z_{m-1})(s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_{n-1})(s - p_n)}$$

z_i : system zeros or roots of the equation $N(s) = 0$

p_i : system poles or roots of the equation $D(s) = 0$

$D(s)=0$ is called the characteristic equation.

All of the coefficients of polynomials $N(s)$ and $D(s)$ are real, therefore the poles and zeros must be either purely real, or appear in complex conjugate pairs.

Transient Response and Stability

• Poles and Zeros

Example-1: A linear system is described by the differential equation

$$\frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y = 2 \frac{du}{dt} + u$$

Find the system poles and zeros.

Solution: Taking L.T. for both sides of the equation gives

$$s^2 Y(s) + 5sY(s) + 6Y(s) = 2sU(s) + U(s)$$

Therefore the transfer function is

$$H(s) = \frac{Y(s)}{U(s)} = \frac{2s + 1}{s^2 + 5s + 6} = \frac{2(s - (-1/2))}{(s - (-2))(s - (-3))}$$

The system therefore has a single real zero at $s = -1/2$, and a pair of real poles at $s = -3$ and $s = -2$.

Transient Response and Stability

• Poles and Zeros

Example-2: A system has a pair of complex conjugate poles $p_1, p_2 = -1 \pm j2$, a single real zero $z_1 = -4$, and a gain factor $k = 3$. Find the differential equation representing the system.

Solution: The transfer function is

$$\begin{aligned} H(s) &= K \frac{s - z}{(s - p_1)(s - p_2)} = 3 \frac{s - (-4)}{(s - (-1 + j2))(s - (-1 - j2))} \\ &= \frac{3s + 12}{s^2 + 2s + 5} \end{aligned}$$

and the differential equation is

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 5y = 3 \frac{du}{dt} + 12u$$

Transient Response and Stability

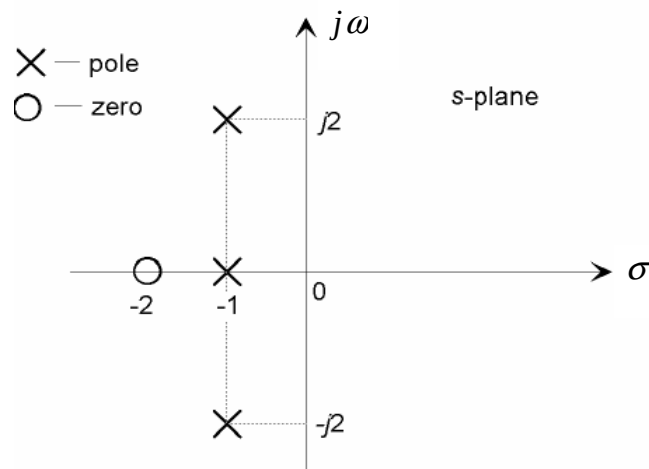
• Poles and Zeros

A system is characterized by its poles and zeros in the sense that they allow reconstruction of the input/output differential equation. In general, the poles and zeros of a transfer function may be complex, and the system dynamics may be represented graphically by plotting their locations on the complex s-plane, whose axes represent the real and imaginary parts of the complex variable s. Such plots are known as **pole-zero plots**. The location of the poles and zeros provide qualitative insights into the response characteristics of a system.

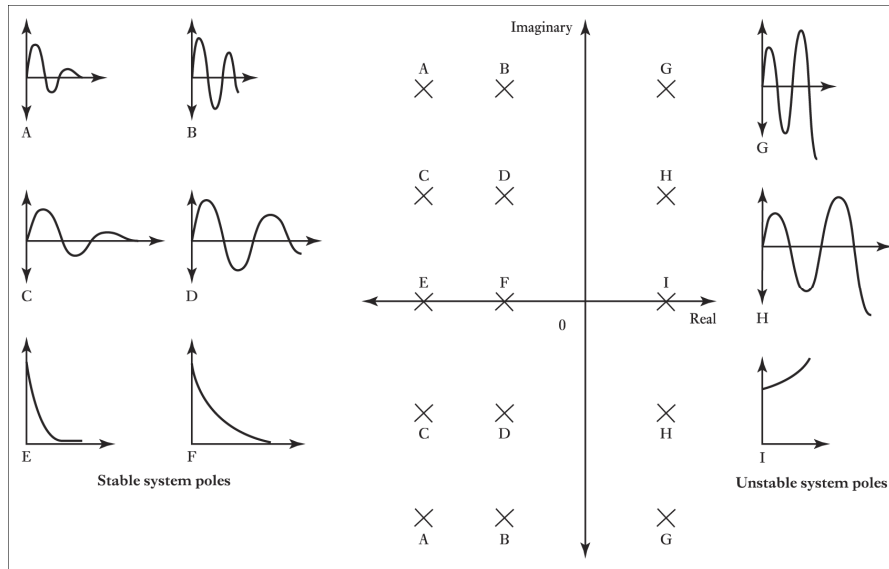
Transient Response and Stability

• Poles and Zeros

$$H(s) = \frac{(3s+6)}{(s^3+3s^2+7s+5)} = 3 \frac{(s-(-2))}{(s-(-1))(s-(-1-2j))(s-(-1+2j))}$$



Transient Response and Stability



The s-Plane

Transient Response and Stability

• Simple Lag: First-Order Systems

Transfer functions in the separate form:

$$H(s) = \frac{N(s)}{D(s)} = K \frac{(s - z_1)(s - z_2) \dots (s - z_{m-1})(s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_{n-1})(s - p_n)}$$

$$\mathcal{L}^{-1} \left\{ H(s) = \frac{K_1}{s - p_1} + \frac{K_2}{s - p_2} + \dots + \frac{K_n}{s - p_n} \right.$$

$$\left. \rightarrow h(t) = K_1 e^{p_1 t} + K_2 e^{p_2 t} + \dots + K_n e^{p_n t} \right.$$

K_i are the residues of $H(s)$ at the corresponding poles.

$$K_i = \left[(s - p_i) H(s) \right]_{s=p_i}$$

Note:

$$Ae^{-\alpha t} \xleftrightarrow{\mathcal{L}} \frac{A}{s + \alpha} \xleftrightarrow{\mathcal{L}^{-1}} Ae^{-\alpha t}$$

Transient Response and Stability

• Simple Lag: First-Order Systems

Transfer functions in the separate form:

Example:

Given
$$F(s) = \frac{s+2}{s(s+1)(s+3)}$$

Required: $f(t)$

Solution:
$$F(s) = \frac{s+2}{s(s+1)(s+3)} = \frac{K_0}{s} + \frac{K_1}{s+1} + \frac{K_2}{s+3}$$

$$K_0 = [sF(s)]_{s=0} = \left[\frac{s+2}{(s+1)(s+3)} \right]_{s=0} = 2/3$$

$$K_1 = [(s+1)F(s)]_{s=-1} = \left[\frac{s+2}{s(s+3)} \right]_{s=-1} = -1/2$$

Transient Response and Stability

• Simple Lag: First-Order Systems

Transfer functions in the separate form:

Example (cont'd):

$$K_2 = [(s+3)F(s)]_{s=-3} = \left[\frac{s+2}{s(s+1)} \right]_{s=-3} = -1/6$$

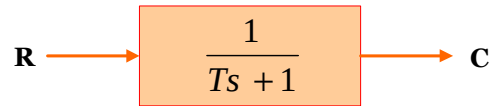
$$F(s) = \frac{2/3}{s} - \frac{1/2}{s+1} - \frac{1/6}{s+3}$$

Taking inverse L.T. gives

$$f(t) = \frac{2}{3} - \frac{1}{2}e^{-t} - \frac{1}{6}e^{-3t}$$

Transient Response and Stability

• Simple Lag: First-Order Systems



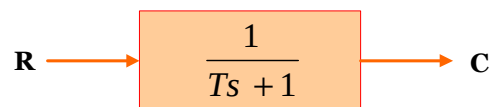
$$C(s) = \frac{1/T}{s(s + 1/T)} = \frac{K_1}{s} + \frac{K_2}{s + 1/T}$$

$$K_1 = [sC(s)]_{s=0} = s \frac{1/T}{s(s + 1/T)} \Big|_{s=0} = 1$$

$$K_2 = [(s + 1/T)C(s)]_{s=-1/T} = (s + 1/T) \frac{1/T}{s(s + 1/T)} \Big|_{s=-1/T} = -1$$

Transient Response and Stability

• Simple Lag: First-Order Systems



$$C(s) = \frac{1}{s} - \frac{1}{s + 1/T}$$

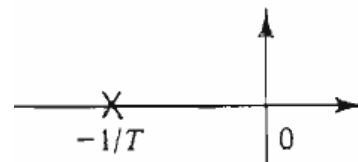
Hence the transient response is

$$c(t) = 1 - e^{-t/T}$$

The forced solution, due to the input



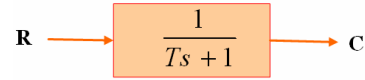
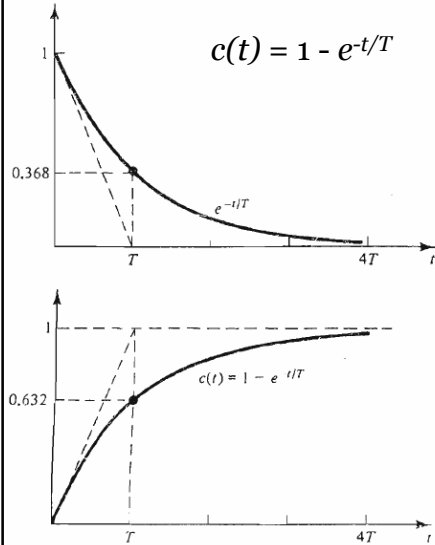
The transient solution, due to the system pole.



Simple Lag

Transient Response and Stability

• Simple Lag: First-Order Systems



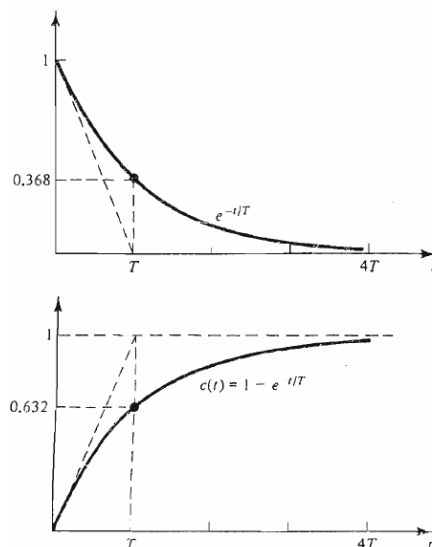
Note that: The transient is seen to be a decaying exponential. If it takes long to decay, the system response is slow, so the speed of decay is of key importance. The commonly used measure of this speed of decay is the time constant.

Step response of simple lag

Transient Response and Stability

• Simple Lag: First-Order Systems

- ◇ **Time constant T:** This is the time in seconds for the decaying exponential transient to be reduced to $e^{-1} = 0.368$ of its initial value.
- ◇ **T** indicates the speed of decay.
- ◇ It takes $4T$ seconds for the transient to decay to 1.8% of its initial value.



Transient Response and Stability

• Simple Lag: First-Order Systems

For the simple lag, two features are important:

1. **Stability:** As discussed, for stability, the system pole $-1/T$ must lie in the left half of the s-plane, since otherwise the transient $e^{-t/T}$ grows instead of decays as t increases.
2. **Speed of response:** To speed up the response of the system (that is, to reduce its time constant T), the pole $-1/T$ must be moved left.

Transient Response and Stability

• Quadratic Lag: Second-Order Systems

This very common transfer function can always be reduced to the standard form:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

where ω_n = undamped natural frequency
 ζ = damping ratio

For a unit step input $R(s) = 1/s$, the transform of the output is

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

Transient Response and Stability

• Quadratic Lag: Second-Order Systems

There are three possibilities, depending on the roots of the system characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

These system poles depend on ζ :

$\zeta > 1$:

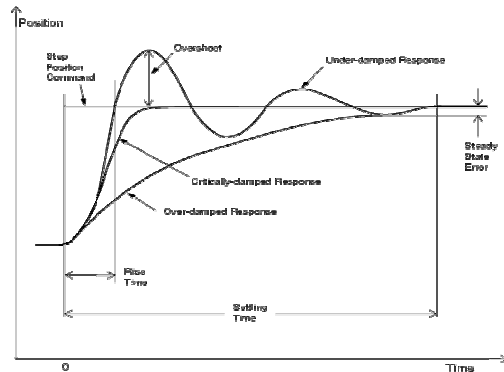
overdamped : $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

$\zeta = 1$:

critically damped : $s_{1,2} = -\omega_n$

$\zeta < 1$:

underdamped : $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$



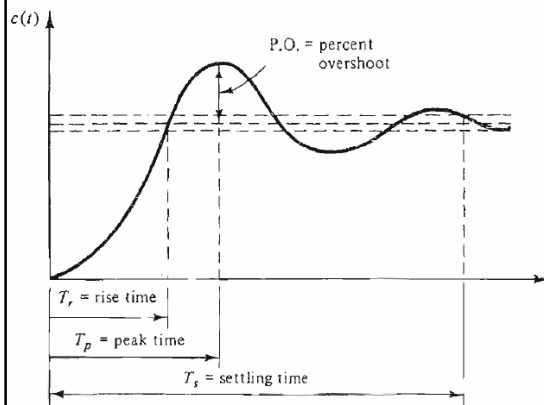
Transient Response and Stability

• Quadratic Lag: Second-Order Systems

$\zeta > 1$: overdamped : $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

$\zeta = 1$: critically damped : $s_{1,2} = -\omega_n$

$\zeta < 1$: underdamped : $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$



$$T = \frac{1}{\zeta\omega_n}$$

$$T_s = 4T = \frac{4}{\zeta\omega_n}$$

$$T_p = \frac{\pi}{\omega_n\sqrt{1 - \zeta^2}}$$

$$P.O. = 100 \exp\left(\frac{-\pi\zeta}{\sqrt{1 - \zeta^2}}\right)$$

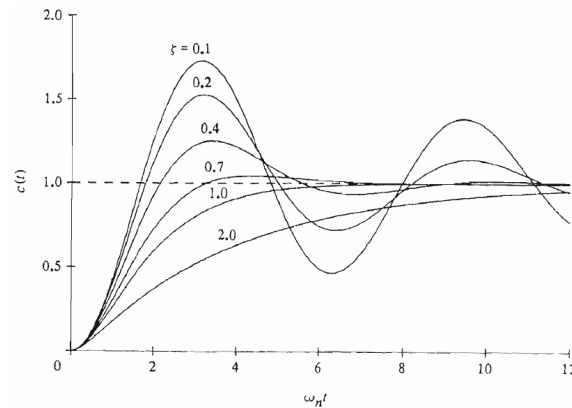
Transient Response and Stability

• Quadratic Lag: Second-Order Systems

$\zeta > 1$: overdamped : $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

$\zeta = 1$: critically damped : $s_{1,2} = -\omega_n$

$\zeta < 1$: underdamped : $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

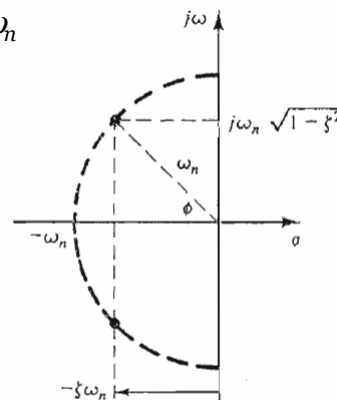


Transient Response and Stability

• Quadratic Lag: Second-Order Systems

As for the simple lag, it is again important to determine the correlations between dynamic behavior and the pole positions in the s-plane

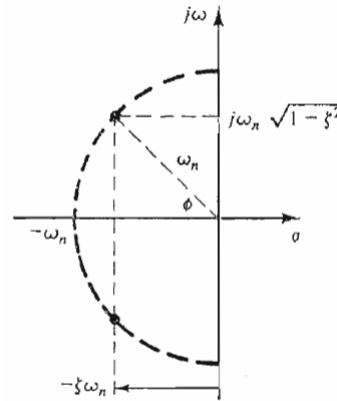
- ◇ **Absolute stability:** The real part $-\zeta\omega_n$ of the poles must be negative for the transient to decay; that is, the poles must lie in the left half of the s-plane.
- ◇ **Relative stability:** To avoid excessive overshoot and unduly oscillatory behavior, damping ratio ζ must be adequate. Since $\zeta = \cos \phi$, the angle ϕ may not be close to 90° .



Transient Response and Stability

• Quadratic Lag: Second-Order Systems

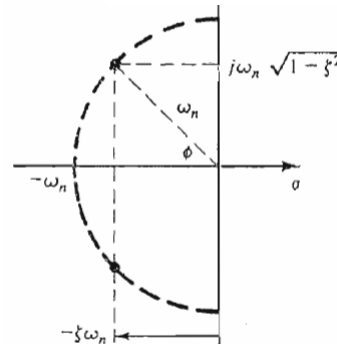
- ◇ **Time constant:** The time constant is reduced (that is, the speed of decay of the transient is increased) by increasing the negative real part of the pole positions.
- ◇ **Speed of response:** Noting that the horizontal axis in terms of ω_n , it is seen that the speed of response of the system is increased by increasing the distance ω_n of the poles to the origin.



Transient Response and Stability

• Quadratic Lag: Second-Order Systems

- ◇ **Undamped natural frequency ω_n :** This equals the distance of the poles to the origin. Moving the poles out radially (that is, with ζ constant) increases the speed of response while the percentage overshoot remains constant.
- ◇ **Frequency of transient oscillations $\omega_n \sqrt{1 - \zeta^2}$:** This frequency, also called the resonant frequency or damped natural frequency, equals the imaginary part of the pole positions.



Transient Response and Stability

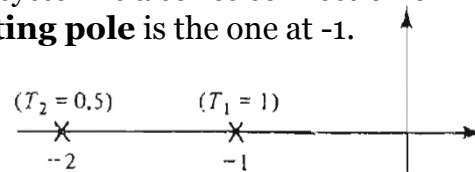
• Quadratic Lag: Second-Order Systems

Example for $\zeta > 1$: $G(s) = \frac{2}{s^2 + 3s + 2} = \frac{2}{(s+1)(s+2)}$

For a unit step, $C(s) = \frac{2}{s(s+1)(s+2)} = \frac{K_1}{s} + \frac{K_2}{s+1} + \frac{K_3}{s+2}$

and it is found that $c(t) = 1 - 2e^{-t} + e^{-2t}$

The transient consists of exponentials with time constants $T_1=1$ and $T_2=0.5$, could have been predicted from inspection of the system pole-zero pattern. The system is a series connection of two simple lags. The **dominating pole** is the one at -1.



Transient Response and Stability

• Higher-Order Systems and Dominating Poles

The *total transient response* is a **superposition** of exponential decays and decaying oscillations.

- ◇ The **dominating-poles** concept simplifies both analysis and design greatly.
- ◇ The response of many systems is dominated by one pair of complex poles relatively close to the imaginary axis, so they behave approximately as **second-order systems**.
- ◇ The number of additional poles in near-dominant locations that can cause deviations from this second-order behavior is usually very limited.

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Absolute stability requires only that all roots of the system characteristic equation

$$a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0$$

(that is, the poles of its transfer function) lie in the left-half s-plane.

Routh-Hurwitz criterion provides a quick method for determining absolute, but not relative, stability from the coefficients, without calculating the roots. It shows how many system poles are in the right-half s-plane or on the imaginary axis.

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

The *Routh-Hurwitz criterion* states:

1. A necessary and sufficient condition for stability is that there be no changes of sign in the elements of the first column of the array.
2. The number of these sign changes is equal to the number of roots in the right-half s-plane.
3. If the first element in a row is zero, it is replaced by a very small positive number ε , and the sign changes when $\varepsilon \rightarrow 0$ are counted after completing the array.
4. If all elements in a row are zero, the system has poles in the right half-plane or on the imaginary axis.

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Routh Array:

$$a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0 \quad \Rightarrow$$

$s^n:$	a_n	a_{n-2}	a_{n-4}	$\dots$	0
	$\downarrow$	$\nearrow$	$\downarrow$	$\nearrow$	
$s^{n-1}:$	a_{n-1}	a_{n-3}	a_{n-5}	$\dots$	0
$s^{n-2}:$	b_1	b_2	b_3	$\dots$	0
$s^{n-3}:$	c_1	c_2	c_3	$\dots$	0
$\vdots$					
$s^1:$	g_1	0			
$s^0:$	h_1	0			

$$b_1 = \frac{1}{a_{n-1}} (a_{n-1} a_{n-2} - a_n a_{n-3})$$

$$b_2 = \frac{1}{a_{n-1}} (a_{n-1} a_{n-4} - a_n a_{n-5})$$

$$b_3 = \frac{1}{a_{n-1}} (a_{n-1} a_{n-6} - a_n a_{n-7})$$

$$c_1 = \frac{1}{b_1} (b_1 a_{n-3} - b_2 a_{n-1})$$

$$c_2 = \frac{1}{b_1} (b_1 a_{n-5} - b_3 a_{n-1})$$

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Example-1: $s^5 + s^4 + 6s^3 + 5s^2 + 12s + 20 = 0$

Routh array is

$$s^5: 1 \quad 6 \quad 12 \quad 0$$

$$s^4: 1 \quad 5 \quad 20 \quad 0$$

$$s^3: 1 \quad -8 \quad 0$$

$$s^2: 13 \quad 20 \quad 0$$

$$s^1: \frac{-124}{13} \quad 0$$

$$s^0: 20$$

There are two sign changes in the first column, first from plus to minus and then from minus to plus. Hence this characteristic equation represents an unstable system, with two poles in the right-half s-plane

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Example-2: $s^5 + s^4 + 5s^3 + 5s^2 + 12s + 10 = 0$

$s^5:$	1	5	12	0
$s^4:$	1	5	10	0
$s^3:$	ε	2	0	
$s^2:$	c_1	10	0	
$s^1:$	d_1	0		
$s^0:$	10			

The zero element in the first column for s^3 in the following Routh array is replaced by a small positive number ε . Then

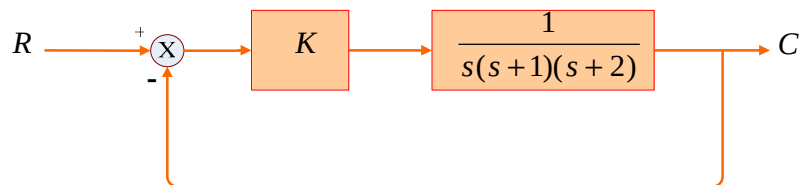
$$c_1 = \frac{5\varepsilon - 2}{\varepsilon} \approx \frac{-2}{\varepsilon} \quad d_1 = \frac{2c_1 - 10\varepsilon}{c_1} \approx 2$$

Thus c_1 is a large negative number, implying two unstable poles

Transient Response and Stability

• Higher-Order Systems and Dominating Poles

Example-3: For the electromechanical servo modeled in the following diagram, find the limits on K for stability.



$$\frac{C}{R} = \frac{K}{s^3 + 3s^2 + 2s + K}$$

Transient Response and Stability

- Higher-Order Systems and Dominating Poles

Example-3 (cont'd):

$$\frac{C}{R} = \frac{K}{s^3 + 3s^2 + 2s + K}$$

The characteristic equation is $s^3 + 3s^2 + 2s + K = 0$

Routh array is

s^3	:	1	2	0
s^2	:	3	K	0
s^1	:	$\frac{6-K}{3}$	0	
s^0	:	K	0	

Evidently, $0 < K < 6$ is the range of K for a stable system.

Outline

- Introduction
- Transient Response and Stability
- **Transient Response vs Steady-State Errors**
- Summary

Transient Response vs Steady-State Errors

- ◇ It has been shown that parameter sensitivity, disturbance response, and steady-state errors are all improved by increased gain.
- ◇ Dynamic response considerations usually limit the permissible gain as **transient response and accuracy** are related.
- ◇ A **higher gain** means a larger signal to the actuator for a certain change of error. This suggests smaller error, but also increasing **danger of overcorrection and oscillations**.
- ◇ In fact, feedback design is generally concerned with achieving a **satisfactory compromise** between relative stability, that is, **adequate damping ratio** ζ of the dominating closed-loop poles, and accuracy, that is, **small errors**.

Transient Response vs Steady-State Errors

Example-1: Step Responses of a DC Motor Position Servo

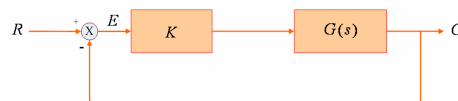
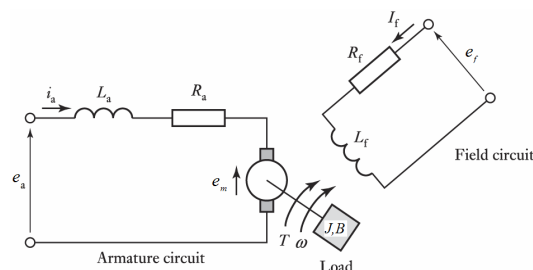
let the transfer function from field voltage to shaft position of a field-controlled dc motor be

$$G(s) = \frac{0.5}{s(0.25s + 1)}$$

Assuming a fast response power amplifier and a constant-gain controller, the loop can be considered to be closed, as shown, by an amplifier block of gain K .

The closed-loop transfer function is:

$$\frac{C(s)}{R(s)} = \frac{2K}{s^2 + 4s + 2K}$$



Transient Response vs Steady-State Errors

Example-1: Step Responses of a DC Motor Position Servo

and for a unit step input [$R(s) = 1/s$], the transform of the shaft position is:

$$C(s) = \frac{2K}{s(s^2 + 4s + 2K)}$$

- $K = 1$

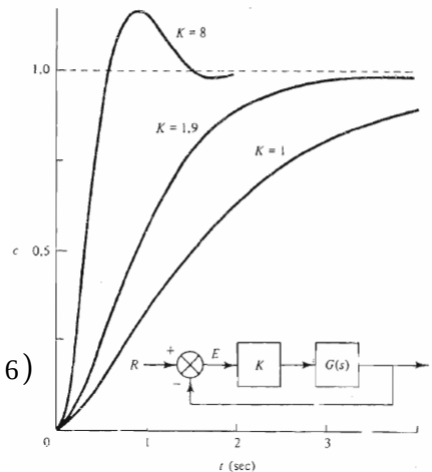
$$c(t) = 1 - 1.207 e^{-0.586 t} + 0.207 e^{-3.414 t}$$

- $K = 1.9$

$$c(t) = 1 - 2.737 e^{-1.553 t} + 1.737 e^{-2.447 t}$$

- $K = 8$

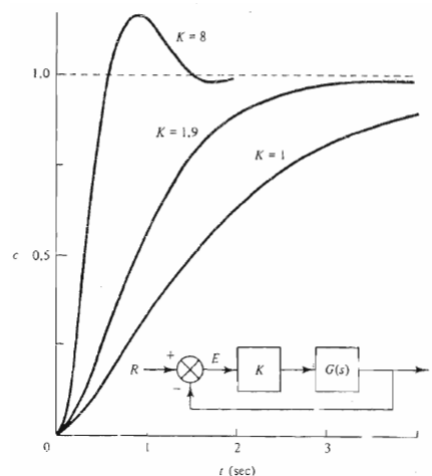
$$c(t) = 1 + 1.1547 e^{-2t} \cos(3.4641 t + 5\pi / 6)$$



Transient Response vs Steady-State Errors

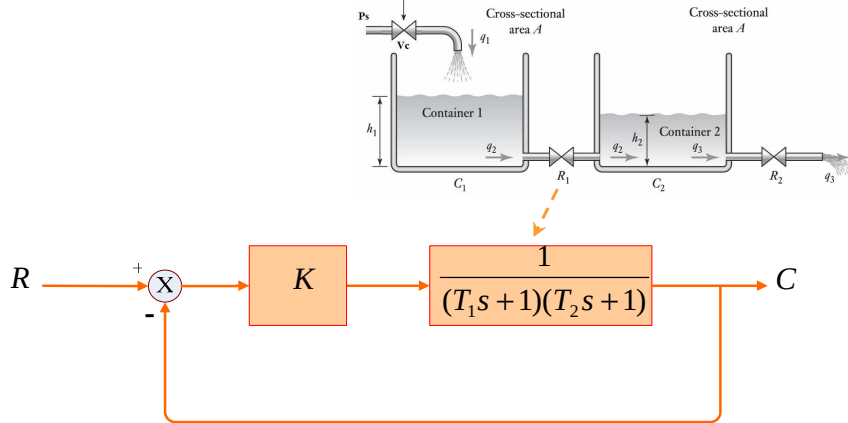
Example-1: Step Responses of a DC Motor Position Servo

- ◇ **Larger K produces faster response** for, because the motor input changes more for a given change of error, and a danger of **overcorrection and oscillations** if K is raised further.
- ◇ The steady-state or final value of the error is seen to be zero for all three values of K, as expected for this **type 1** system.



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Example-2: P-Control with Two Simple Lags



Among other possibilities, this system could represent a two-tank level control, a temperature control with two heat capacitances in series, or a motor speed control.

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Example-2: P-Control with Two Simple Lags

the system is **type 0** with gain K , so the steady-state error after a unit step is $1/(1 + K)$

Consider now stability, using $T_1 = 1$, $T_2 = 0.5$ for a numerical example. The closed-loop transfer function and the system characteristic equation are, respectively,

$$\frac{C(s)}{R(s)} = \frac{2K}{s^2 + 3s + 2 + 2K}$$

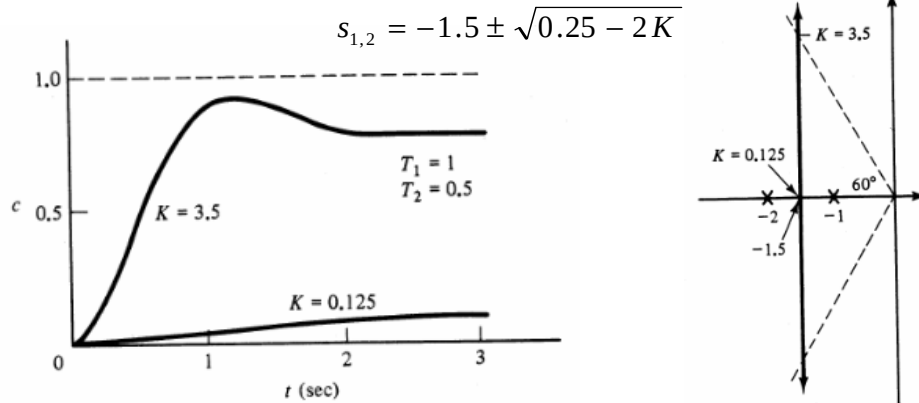
$$\text{Characteristic equations: } s^2 + 3s + 2 + 2K = 0$$

The closed-loop poles are:

$$s_{1,2} = -1.5 \pm \sqrt{0.25 - 2K}$$

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Example-2: P-Control with Two Simple Lags



The increase of K that is desirable to increase the speed of response and to reduce steady-state errors, parameter sensitivity, and disturbance response is limited by considerations of relative stability.

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- ◇ Often the maximum value of K permitted by these considerations is not large enough to meet specifications on the steady-state error $1/(1 + K)$.
- ◇ Indeed, in many cases this error is required to be zero. This motivates the use of dynamic compensation.
- ◇ In proportional control, or P control, only gain adjustment is available to improve performance.
- ◇ If performance specifications cannot be met by P control, it must be replaced by a **dynamic controller or dynamic compensator** to provide more flexibility.

Outline

- Introduction
- Transient Response and Stability
- Transient Response vs Steady-State Errors
- **Summary**

Summary

- Considerations of relative stability usually limit the gain increase, which is desirable to increase the speed of response and to reduce the sensitivity to plant parameter variations and model approximations, as well as to reduce errors and the response to disturbances.
- The dynamic response characteristics have been interpreted in terms of the locations of system poles in the s-plane. These poles must lie in the left half-plane to ensure stability. The Routh-Hurwitz criterion permits this to be determined without actual calculation of the poles.

Questions?